Notation

$\bar{w} = h/R$	true height of the particle centre, units of R
$\bar{w}_d,\ \Delta\bar{w}_d[k] = \bar{w}_d[k+1] - \bar{w}_d[k]$	commanded height and commanded step
$\delta\bar{w}_i$	tracking deviation, $\bar{w} = \bar{w}_d - \delta\bar{w}_3$; $\delta\bar{w}_1$ = the same two steps later ($d = 2$)
$\bar{a} = a/a_{\mathrm{nom}}$	gain, normalised by the far-field gain
$\bar{a}' = d\bar{a}/d\bar{w},\ \bar{a}'' = d^2\bar{a}/d\bar{w}^2$	slope and curvature of the gain in height
b	the law's constant; $b_{\mathrm{true}}(\bar{w})$ the exact factor of the true curve
$A = \partial\bar{a}'/\partial\bar{a}$	self-sensitivity of the slope
$\hat{\theta}$	estimator's value of θ; $\bar{a}',\ \bar{a}'',\ A$ without subscript are evaluated at $(\hat{\bar{a}}_w, \hat{b})$
$e_\theta = \theta - \hat{\theta}$	error, TRUE minus ESTIMATOR (SSOT convention)
$\Delta\bar{w},\ \Delta\hat{\bar{w}},\ e_{\Delta\bar{w}} = \Delta\bar{w} - \Delta\hat{\bar{w}}$	true step, estimator's step, step error
F_{dw}	loop coupling: the normalised force the controller applied (previous step)
ε_w	thermal step of the position (variance Q_{33})
$n_w,\ n_a$	position and gain-readout measurement noise (R_1, R_2)
λ_c	pole of the position loop
$\nabla_d\bar{w}_d$	commanded displacement the delayed readout walks back
$P_{ij} = \mathbb{E}[e_i e_j]$	error covariance; F_e, H, Q its propagation

1 Gain model

Law, its derivatives and its solution (the estimator's model; $\hat{b}$ constant)

$$\bar{a}' = b\,(1-\bar{a})^2$$

$$A = \frac{\partial\bar{a}'}{\partial\bar{a}} = -2\,b\,(1-\bar{a})$$

$$\frac{\partial\bar{a}'}{\partial b} = (1-\bar{a})^2 = \frac{\bar{a}'}{b}$$

$$\bar{a}'' = \frac{d\bar{a}'}{d\bar{w}} = A\,\bar{a}' = -2\,b\,(1-\bar{a})\,\bar{a}'$$

$$\frac{1}{1-\bar{a}(\bar{w})} = b\,(\bar{w}-\bar{w}_s) \qquad \Longleftrightarrow \qquad \bar{a}(\bar{w}) = 1 - \frac{1}{b\,(\bar{w}-\bar{w}_s)}$$

True curve (published plane: $b_{\mathrm{true}}(\bar{w})$ a function, $0.883 \to 0.867$ at $\bar{w} = 2.19 \to 0.928$ at $\bar{w} = 1.10$)

$$\bar{a}'_{\mathrm{true}}(\bar{w}) = b_{\mathrm{true}}(\bar{w})\,\big(1-\bar{a}_{\mathrm{true}}(\bar{w})\big)^2$$

$$\bar{a}''_{\mathrm{true}}(\bar{w}) = b'_{\mathrm{true}}(\bar{w})\,\big(1-\bar{a}_{\mathrm{true}}\big)^2 - 2\,b_{\mathrm{true}}(\bar{w})\,\big(1-\bar{a}_{\mathrm{true}}\big)\,\bar{a}'_{\mathrm{true}}$$

Model error of a constant $\hat{b}$ (the only approximation of the law), and the true slope and curvature written around the estimator's point

$$e_b[k] := b_{\mathrm{true}}\big(\bar{w}[k]\big) - \hat{b}[k]$$

$$f[k] := \big(1-\hat{\bar{a}}_w[k]\big)^2\,e_b[k]$$

$$\bar{a}'_{\mathrm{true}}\big(\bar{w}[k]\big) = \bar{a}' + A\,e_{\bar{a}_w}[k] + f[k] + \hat{b}\,e_{\bar{a}_w}[k]^2 - 2\,(1-\hat{\bar{a}}_w)\,e_{\bar{a}_w}[k]\,e_b[k] + e_b[k]\,e_{\bar{a}_w}[k]^2 \qquad \text{(exact)}$$

$$\bar{a}''_{\mathrm{true}}\big(\bar{w}[k]\big) = \bar{a}'' + f'[k] + \frac{\bar{a}''}{\bar{a}'}\,f[k] + \mathcal{O}(e), \qquad f' = \frac{df}{d\bar{w}}$$

2 True dynamics

$$
\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
\bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'_{\text{true}}\big(\bar{w}[k]\big)\,\Delta\bar{w}[k] + \tfrac{1}{2}\,\bar{a}''_{\text{true}}\big(\bar{w}[k]\big)\,\Delta\bar{w}[k]^2 + \mathcal{O}\big(\Delta\bar{w}^3\big) \\
b_{\text{true}}\big(\bar{w}[k+1]\big) &= b_{\text{true}}\big(\bar{w}[k]\big) + b'_{\text{true}}\big(\bar{w}[k]\big)\,\Delta\bar{w}[k] + \mathcal{O}\big(\Delta\bar{w}^2\big) \\
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k] - \bar{a}'_{\text{true}}\big(\bar{w}[k]\big)\,\nabla_d\bar{w}_d[k] + n_a[k]
\end{aligned}
$$

3 Estimator

State $\hat{x} = [\,\delta\hat{\bar{w}}_1,\ \delta\hat{\bar{w}}_2,\ \delta\hat{\bar{w}}_3,\ \hat{\bar{a}}_w,\ \hat{b}\,]^\top$; predict

$$
\begin{aligned}
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\Delta\hat{\bar{w}}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + \bar{a}'\,\Delta\hat{\bar{w}}[k] + \tfrac{1}{2}\,\bar{a}''\,\Delta\hat{\bar{w}}[k]^2 + m_2[k] \\
\hat{b}[k+1] &= \hat{b}[k]
\end{aligned}
$$

($m_2[k]$ = the mean of the second-order error group, Section 4, added back so that $\mathbb{E}[e_{\bar{a}_w}]$ has no drift; the exact-step form $1/(1-\hat{\bar{a}}_w[k+1]) = 1/(1-\hat{\bar{a}}_w[k]) + \hat{b}\,\Delta\hat{\bar{w}}[k]$ replaces the two law terms in the code and agrees with them to second order)

Predicted measurements

$$
\begin{aligned}
\hat{y}_1[k] &= \delta\hat{\bar{w}}_1[k] \\
\hat{y}_2[k] &= \hat{\bar{a}}_w[k] - \bar{a}'\,\nabla_d\bar{w}_d[k]
\end{aligned}
$$

4 Error dynamics

Step error and its moments ($u := e_{\Delta\bar{w}}$)

$$
\begin{aligned}
e_{\Delta\bar{w}}[k] &= \Delta\bar{w}[k] - \Delta\hat{\bar{w}}[k] = (1-\lambda_c)\,e_{\delta\bar{w}_3}[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
\mathbb{E}[e_{\Delta\bar{w}}] &= 0, \qquad \operatorname{Var}(e_{\Delta\bar{w}}) = (1-\lambda_c)^2 P_{33} + F_{dw}^2 P_{44} + 2(1-\lambda_c)F_{dw}P_{34} + Q_{33} \\
\operatorname{Cov}(e_{\delta\bar{w}_3}, e_{\Delta\bar{w}}) &= (1-\lambda_c)P_{33} + F_{dw}P_{34} \\
\operatorname{Cov}(e_{\bar{a}_w}, e_{\Delta\bar{w}}) &= (1-\lambda_c)P_{34} + F_{dw}P_{44} \\
\operatorname{Cov}(e_b, e_{\Delta\bar{w}}) &= (1-\lambda_c)P_{35} + F_{dw}P_{45}
\end{aligned}
$$

Position chain (exact)

$$
\begin{aligned}
e_{\delta\bar w_1}[k+1] &= e_{\delta\bar w_2}[k] \\
e_{\delta\bar w_2}[k+1] &= e_{\delta\bar w_3}[k] \\
e_{\delta\bar w_3}[k+1] &= \lambda_c\, e_{\delta\bar w_3}[k] - F_{dw}[k]\, e_{\bar a_w}[k] - \varepsilon_w[k]
\end{aligned}
$$

Gain, all terms to second order (true increment minus estimator increment, $\Delta\bar w = \Delta\hat{\bar w} + e_{\Delta\bar w}$)

$$
\begin{aligned}
e_{\bar a_w}[k+1] - e_{\bar a_w}[k] &= \Big[\bar a' + A\, e_{\bar a_w} + f + \hat b\, e_{\bar a_w}^2 - 2(1-\hat{\bar a}_w) e_{\bar a_w} e_b + e_b e_{\bar a_w}^2\Big]\big(\Delta\hat{\bar w} + e_{\Delta\bar w}\big) \\
&\quad + \tfrac{1}{2}\Big[\bar a'' + f' + \tfrac{\bar a''}{\bar a'} f\Big]\big(\Delta\hat{\bar w} + e_{\Delta\bar w}\big)^2 - \bar a'\,\Delta\hat{\bar w} - \tfrac{1}{2}\,\bar a''\,\Delta\hat{\bar w}^2 - m_2 \\
&= \underbrace{\bar a'\, e_{\Delta\bar w} + A\,\Delta\hat{\bar w}\, e_{\bar a_w} + \Delta\hat{\bar w}\, f}_{(1)\ \text{first order}} + \underbrace{A\, e_{\bar a_w}\, e_{\Delta\bar w} + f\, e_{\Delta\bar w} + \bar a''\,\Delta\hat{\bar w}\, e_{\Delta\bar w} + \tfrac{1}{2}\,\bar a''\, e_{\Delta\bar w}^2}_{(2)\ \text{second order, the } \bar a'' \text{ group}} \\
&\quad + \underbrace{\tfrac{1}{2}\Big(f' + \tfrac{\bar a''}{\bar a'} f\Big)\big(\Delta\hat{\bar w} + e_{\Delta\bar w}\big)^2 + \hat b\, e_{\bar a_w}^2\big(\Delta\hat{\bar w} + e_{\Delta\bar w}\big) - 2(1-\hat{\bar a}_w)\, e_{\bar a_w} e_b\big(\Delta\hat{\bar w} + e_{\Delta\bar w}\big)}_{(3)\ \text{second order, model-error curvature and squares}} - m_2 + e_b\, e_{\bar a_w}^2
\end{aligned}
$$

(grouping rule: every term that carries a curvature, $\bar a''$ or f', is counted in (2)/(3) whatever its degree in the small quantities; group (1) is exactly the degree-1 terms without curvature. The last term is third order. Independent sympy verification 09-07: groups (1)–(3) reproduce the difference of the two increments term by term; $e_b = 0$, $f' = 0$ returns the $\mathrm{a}'_{\text{true}}$-arm expressions of the SSOT exactly)

Law constant (the constant model has no term for the second line)

$$
e_b[k+1] = e_b[k] + b'_{\text{true}}\big(\bar w[k]\big)\,\Delta\bar w[k]
$$

Mean of the second-order group (this is m_2; $\mathbb{E}[e] = 0$ for every state, $\mathbb{E}[e_{\Delta\bar w}] = 0$)

$$
\begin{aligned}
m_2[k] &= A\,\mathrm{Cov}(e_{\bar a_w}, e_{\Delta\bar w}) + \tfrac{1}{2}\,\bar a''\,\mathrm{Var}(e_{\Delta\bar w}) + \frac{\bar a' + \bar a''\,\Delta\hat{\bar w}}{\hat b}\,\mathrm{Cov}(e_b, e_{\Delta\bar w}) \\
&\quad + \tfrac{1}{2}\, f'\big(\Delta\hat{\bar w}^2 + \mathrm{Var}(e_{\Delta\bar w})\big) + \hat b\, P_{44}\,\Delta\hat{\bar w} - 2(1-\hat{\bar a}_w)\, P_{45}\,\Delta\hat{\bar w} + (\text{third moments})
\end{aligned}
$$

($\mathbb{E}[e_b] = 0$ kills the $\Delta\hat{\bar w}^2$ part of the $\frac{\bar a''}{\bar a'} f$ line but not its cross term $2\Delta\hat{\bar w}\, e_{\Delta\bar w}$, which is the $\bar a''\Delta\hat{\bar w}/\hat b$ share of the $\mathrm{Cov}(e_b, e_{\Delta\bar w})$ coefficient – caught by the sympy verification; the code's pred_mean2_e4 e_b line carries only $\bar a'/\hat b$, short by the factor $1 + A\Delta\hat{\bar w}$, i.e. under 1% per step. Code: pred_mean2 = the first two terms, pred_mean2_e4 = the $A\,\mathrm{Cov}(e_{\bar a_w}, e_{\Delta\bar w})$ form and the e_b line; f' needs b_{true} and is NOT in the estimator; the two square terms are $\mathcal{O}(P)\,\Delta\hat{\bar w}$ and are dropped)

First-order part written as $e[k+1] = F_e\, e[k] + g\,\varepsilon_w[k] + (\text{model error})$, $e = [e_{\delta\bar w_1},\ e_{\delta\bar w_2},\ e_{\delta\bar w_3},\ e_{\bar a_w},\ e_b]^\top$

$$
F_e = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 \\
0 & 0 & \lambda_c & -F_{dw} & 0 \\
0 & 0 & (1-\lambda_c)\,\bar a' & 1 + \bar a' F_{dw} + A\,\Delta\hat{\bar w} & \dfrac{\bar a'}{\hat b}\,\Delta\hat{\bar w} \\
0 & 0 & 0 & 0 & 1
\end{bmatrix}, \qquad
g = \begin{bmatrix} 0 \\ 0 \\ -1 \\ \bar a' \\ 0 \end{bmatrix}, \qquad
Q = Q_{33}\, g\, g^\top
$$

$$
\begin{aligned}
\text{row 4}: &\quad \Delta\hat{\bar w}[k]\, f[k] = \Delta\hat{\bar w}[k]\,(1-\hat{\bar a}_w[k])^2\, e_b[k] = F_e(4,5)\, e_b[k] \\
\text{row 5}: &\quad b'_{\text{true}}\big(\bar w[k]\big)\,\Delta\bar w[k]
\end{aligned}
$$

(row 4: the product is already in F_e, but $\mathbb{E}[e_b] \neq 0$ – a curve minus a constant – so it carries a MEAN; row 5: no container in F_e or Q)

Measurement Jacobians ($\partial y/\partial x$ at the estimate)

$$H_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$
$$H_2 = \begin{bmatrix} 0 & 0 & 0 & 1 - A\,\nabla_d \bar{w}_d & -\dfrac{\bar{a}'}{\hat{b}}\,\nabla_d \bar{w}_d \end{bmatrix}, \qquad R = \mathrm{diag}(R_1,\ R_2)$$

Where each entry comes from

$$\begin{aligned}
F_e(3,4) = -F_{dw}: &\quad \text{force computed with } \hat{\bar{a}}_w,\ \text{position moved by } \bar{a}_w \\
F_e(4,3) = (1-\lambda_c)\,\bar{a}': &\quad \partial\Delta\hat{\bar{w}}/\partial\delta\hat{\bar{w}}_3 = 1-\lambda_c \\
F_e(4,4) = 1 + \bar{a}' F_{dw} + A\,\Delta\hat{\bar{w}}: &\quad \bar{a}'\,\partial e_{\Delta\bar{w}}/\partial e_{\bar{a}_w} = \bar{a}' F_{dw}; \quad A\,\Delta\hat{\bar{w}} = \text{the slope reads } \hat{\bar{a}}_w \\
F_e(4,5) = (\bar{a}'/\hat{b})\,\Delta\hat{\bar{w}}: &\quad \partial\bar{a}'/\partial\hat{b} \times \text{step; 0 in a hold and at a turning point} \\
H_2(4),\ H_2(5): &\quad \text{the readout walks the slope back over } \nabla_d \bar{w}_d
\end{aligned}$$